

Jinu Jeong, Ph.D.

☎ (217) 778-8644 | ✉ jinu.j0912us@gmail.com | 💼 linkedin.com/in/jinu-jeong | 🌐 frame-research.github.io

SUMMARY

Machine learning researcher with a PhD in Mechanical Engineering and extensive experience developing deep learning models for scientific computing and quantitative finance. Specialized in financial market prediction using AI, such as boosting, neural network, and statistical modeling. Developed forecasting models for financial markets, machine learning methods for molecular dynamics, and high-performance multiphysics simulations, resulting in multiple first-author publications and production-scale research software.

- Physics: Statistical mechanics | Molecular dynamics | Differentiable simulation
- Machine Learning: Object Detection | Transformer | RNN (LSTM) | CNN | Graph Neural
- Programming skills: Python, Pytorch, Tensorflow, Scipy, Scikit-learn | C/C++ | Git | Linux

EDUCATION

University of Illinois at Urbana-Champaign PhD, Mechanical Engineering	Champaign, IL 2019 – 2024
Seoul National University MS, Mechanical Engineering	Seoul, South Korea 2017 – 2019
University of Ulsan BS, Mechanical Engineering	Ulsan, South Korea 2011 – 2017

SELECTED WORK AND RESEARCH EXPERIENCE

Independent Researcher <i>Quantitative Finance and Machine Learning Researcher</i>	Collaborative with UIUC Dept. of Mathematics 2024 – present
--	--

Surge prediction

- Led research on predicting abrupt stock market movements by integrating stochastic differential equation (SDE) models with deep learning approaches.
 - Developed and trained a Neural ODE model with an LSTM encoder, achieving the highest average precision (AP = 0.666) among all evaluated models.
- Trading
- Implemented and benchmarked forecasting models (GBM, jump-diffusion, CNN, LSTM, Transformer, and Neural ODE) on KOSDAQ (1,700+ stocks, 7 years) and S&P 500 (500 stocks, 8 years) datasets, with deep learning models improving AP by approximately 30% over classical SDE baselines.
 - Designed and backtested an automated trading strategy based on model predictions, incorporating realistic market frictions including brokerage fees, taxes, and slippage.

Rivian Automotive Inc. <i>Physics Simulation Researcher</i>	Plymouth, MI 2024 – present
---	--------------------------------

- Led the end-to-end development of a reduced-order noise simulation framework from project conceptualization and algorithm design through implementation and validation, developing reduced-order finite-element algorithms using component mode synthesis and Galerkin projection to achieve over 100× faster computation while maintaining predictive accuracy and enabling automated design iterations.
- Led the development of an automated computational geometry pipeline for parameterized rotor/stator generation, CAD import, geometry healing, and topology validation, designing computational geometry algorithms to eliminate manual preprocessing for downstream finite-element simulations while mentoring an intern throughout development.

University of Illinois at Urbana-Champaign <i>Physics modeling Researcher</i>	Champaign, IL 2019 – 2024
---	------------------------------

- Initiated and led the development of a differentiable Generalized Langevin Equation framework using Neural ODEs for data-driven coarse-grained molecular dynamics.

Other projects

Please visit my portfolio page (<https://sites.google.com/view/jinu-jeong/>) for more information.

SELECTED PUBLICATION

- S. Kim, **J. Jeong**, W. Kang, S. Nam, J. Park, "Predicting Price Surges with Neural ODE," submitted, 2026.
- J. Jeong** and I. Nadkarni, "DiffGLE: Differentiable Coarse-Grained Dynamics using Generalized Langevin Equation," submitted, 2026.
- J. Jeong**, I. Nadkarni, B. Yalcin, and N. R. Aluru, "Modulating Coarse-Grained Dynamics by Perturbing Free Energy Landscapes," *Journal of Physical Chemistry A*, 2024
- J. Jeong**, C. Liang, and N. R. Aluru, "Data-driven molecular dynamics simulation of water isotope separation using a catalytically active ultrathin membrane", *Physical Chemistry Chemical Physics*, 2024.
- J. Jeong**, et al., "Extended DeepILST for Various Thermodynamic States and Applications in Coarse-Graining," *Journal of Physical Chemistry A*, 2022
- J. Jeong**, et al., "Water Isotope Separation with a Single Layer MoS₂ Nanopore," presented at the Society of Engineering Science Annual Meeting, Oct. 16-19, 2022.
- J. Jeong**, et al., "Extended DeepILST: Data-driven Coarse-Graining of Multi-Atom Molecules at Various Thermodynamic States," presented at the 16th U.S. National Congress on Computational Mechanics, Jul. 25-29, 2021.

AWARD AND FELLOWSHIPS

Oden Institute , <i>Fellowship</i>	Aug 2022 – July 2024
Kwanjeong Educational Foundation , <i>Honor Scholarship</i>	Aug 2019 – July 2023
US National Congress on Computational Mechanics (USNCCM16) , <i>Conference Award</i>	July 2021
University of Ulsan , <i>Honor Scholarship</i>	Sep 2014 – May 2016